Line sensor

DETECT | PICO BLOCKS | INPUT

What does it do?

A line sensor tells light and dark surfaces apart underneath a moving project.

An infrared LED shines at the floor and a detector measures how much light bounces back.

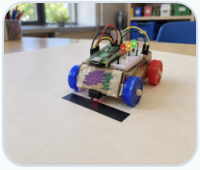
Adjust the small potentiometer until its digital output changes reliably at the line edge.

Uses



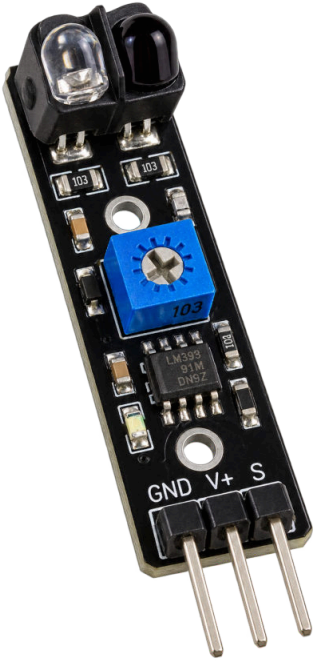
Warehouse guide-line robot

DETECT Read guide-line position → **DECIDE** Is the vehicle drifting? → **DO** Correct the steering



Cardboard lap-counting car

DETECT Cross dark finish strip → **DECIDE** Is this a new lap? → **DO** Increase the lap count



Infrared pair
Faces the floor

Adjuster
Sets the threshold

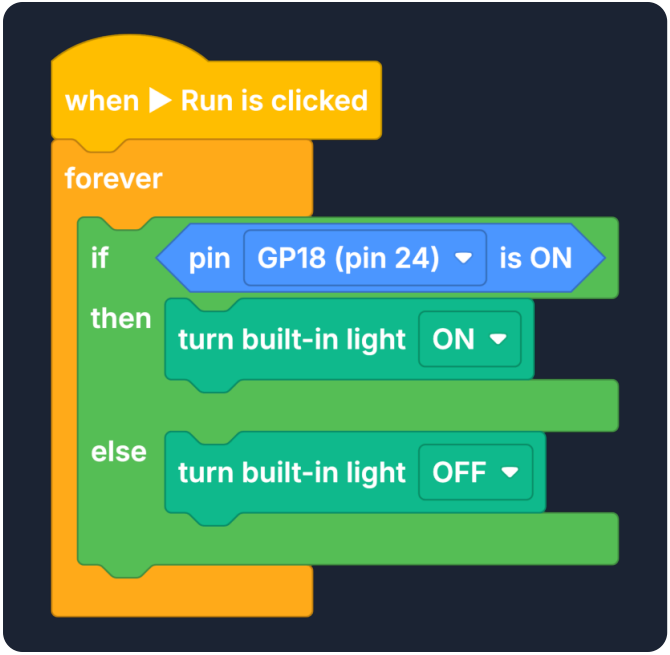
S
Digital signal

Raspberry Pi Foundation

Blocks

MicroLab

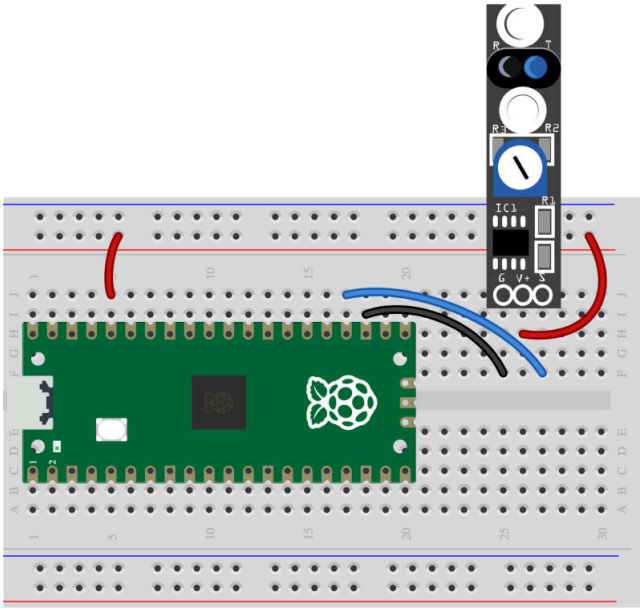
getmicrolab.com



The Pico's light comes on over the dark line. Turn the blue adjuster until it changes cleanly at the line's edge.

Wiring

PICO



GND (pin 23) → G, **3V3 OUT** (pin 36) → V+, **GP18** (pin 24) → S